

Where a Quantum Reservoir Works: A Transferable, Memory-Preserving Operating Band

Markus Baumann^{*} , Itamar Fink^{*}, Johannes Wittmann^{*}, and Jonas Stein^{*} 

^{*}*QAR-Lab, Department of Computer Science, LMU Munich, Munich, Germany*

Corresponding author: markus.baumann@campus.lmu.de

Abstract—Quantum reservoir computing performs temporal machine learning by training only a linear readout on the measured outputs of a fixed quantum system, leaving the quantum dynamics itself untrained. This makes the approach attractive for near-term hardware, but it sharpens a question that has remained largely unresolved: which device settings render the fixed dynamics useful rather than merely high-dimensional. We address this question for a stateful dissipative gate-model reservoir and demonstrate that strong performance does not occupy arbitrary coordinates of the control space. Instead, it concentrates in a well-defined operating region governed by the interplay of input drive, recurrent coupling, and controlled dissipation. This region is transferable: an operating band identified on a subset of tasks predicts effective settings on held-out tasks and held-out reservoir initializations alike. Mechanism ablations establish that the region is causal rather than incidental, and an inexpensive memory-based diagnostic recovers it prior to any task-specific evaluation. The result reframes reservoir selection as the identification of a transferable, memory-preserving regime rather than the pursuit of a single benchmark optimum.

Index Terms—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes

I. INTRODUCTION

Reservoir computing rests on an unusually economical idea: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional trace of its recent history, and learning reduces to fitting a single linear readout on top of that trace [1], [2]. The dynamics are never optimized, only observed. This economy is precisely what makes the approach appealing for quantum hardware, where the native evolution of a system can serve as the reservoir and only the classical readout must be trained [4], [5]. Yet the same economy relocates the entire burden of the method onto the fixed dynamics, and so onto a question that is easy to state and surprisingly hard to answer: what makes those dynamics useful?

Classical reservoir theory has long since refused the obvious answer. Usefulness is not maximal expressivity. A reservoir earns its keep by holding a balance—retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [3], [4], [7]. A quantum reservoir inherits this lesson but complicates it, because the three ingredients are no longer free dials. They are downstream consequences of distinct physical resources: how strongly the input perturbs the carried state, how coherently the system mixes that state across its parts, and how it is allowed to dissipate [4], [6], [8]. A large Hilbert space supplies dimension,

but dimension alone preserves no memory; drive that is too strong overwrites history, and dynamics with no dissipation never forget in the way fading memory requires.

We take this seriously as an organizing principle rather than a caveat. For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask when the reservoir works, and we map it. The picture that emerges is not a single tuned operating point but a connected region—gentle input, nonmaximal coupling, controlled damping—in which good performance concentrates and from which it does not stray arbitrarily across tasks. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and a memory-based diagnostic that locates it before any task-specific search.

II. PROTOCOL

Reservoir. The reservoir is a fixed quantum system that processes a scalar time series one step at a time, carrying its state forward as memory. Concretely, the state is a density matrix that is never reset between steps: each new input perturbs the state left behind by the previous step, so recent history accumulates in the system rather than being overwritten. The input enters as a small rotation whose angle is set by the current value,

$$\rho_{t,0} = U_{\text{in}}(u_t)\rho_{t-1,L}U_{\text{in}}^\dagger(u_t), \quad U_{\text{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \quad (1)$$

The input strength β controls how strongly each step disturbs the carried state.

The perturbed state then evolves through a small number of identical layers. Each layer does three things in turn: it mixes each qubit locally with a frozen rotation, couples neighbors around a ring, and lets the qubits dissipate through a noise channel,

$$\rho_{t,\ell} = \mathcal{D}_\gamma^{\otimes n} \left[U_\lambda U_{\text{mix}} \rho_{t,\ell-1} U_{\text{mix}}^\dagger U_\lambda^\dagger \right], \quad (2)$$

$$U_\lambda = \prod_{(i,j) \in E_{\text{ring}}} e^{-i\lambda Z_i Z_j}.$$

The coupling strength λ sets how much stored history is mixed across qubits, and the damping rate γ sets how quickly the system forgets. These three quantities— β , λ , and γ —are the only controls we vary; every gate is otherwise frozen, and only a linear readout is trained on the measured outputs. Each

control corresponds to one ingredient required by reservoir theory: β governs how new information competes with old memory, λ generates nonlinear mixing of past inputs, and γ provides the controlled forgetting on which fading memory depends. The space $\theta = (\beta, \lambda, \gamma)$ is therefore not an arbitrary grid of knobs but the natural coordinate system in which to ask whether the reservoir is useful.

Configuration. Unless stated otherwise, the reservoir uses four qubits, two layers, ring coupling, uniform input injection, amplitude damping, and a readout built from single-qubit Z and ring-pair ZZ expectation values. Training is ridge regression on these features alone. Hyperparameters are selected on validation data only; holdout data never participate in defining the operating region.

Operating band. To locate the useful region we rank, rather than threshold, performance. For each task j and reservoir seed s , a setting θ receives a validation percentile rank $r_{j,s}(\theta)$, with lower rank meaning better validation error. We then define the operating band as the set of settings that land in the top fraction p for at least a fraction q of all task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \leq p] \geq q\}. \quad (3)$$

This is deliberately a frequency-level definition, weaker than demanding a single universal optimum. It does not ask whether one coordinate wins every task; it asks whether a region behaves well repeatedly—across tasks and across initializations. That distinction is what separates a transferable operating regime from a point that merely happened to win once.

III. EVIDENCE

Across the control space, low validation rank does not spread arbitrarily; it gathers into a compact, connected region, and relaxing the selection threshold grows that region smoothly rather than fragmenting it—the signature of a single operating regime rather than a scatter of coincidences (Fig. 1). A region that merely fit the four tasks jointly would prove little, so we test whether it survives the removal of the information it was built from. It does: when one task is withheld and the band is defined from the rest, the held-out task still lands in the top quartile on average, with mean validation rank 0.240 and CI [0.215, 0.265]; when a seed is withheld instead, the held-out seed behaves the same way, with mean rank 0.180 and CI [0.154, 0.209]. The region is thus a search prior that generalizes across both task identity and reservoir initialization, not a fit to either—and it holds on data it never touched: fixing the band by validation alone and evaluating once on the held-out split, with no coordinate chosen by holdout error, leaves it top-quartile on unseen data (Table II). That the band is real rather than cosmetic is confirmed by perturbing the dynamics that sustain it: removing damping, removing recurrent mixing, or replacing amplitude damping with dephasing each erases the region entirely, with retention values 0.00, 0.00, and 0.00 under the same criterion. The collapse is total rather than gradual, exactly as expected if the regime is held in place by the mechanisms themselves—input strong enough to write but not overwrite, mixing that builds nonlinear features, and damping

that preserves usable memory rather than merely adding noise (Fig. 2). And the region can be found before any task is run at all: memory capacity correlates strongly and negatively with validation rank, $\rho_s = -0.85$, as do the information-processing capacities [3], whereas raw feature diversity does not, telling us that what marks a useful reservoir is not how much it expresses but how much it remembers (Fig. 3).

Variant	Band pts.	Retention	Rank on base band
base AD+RxZZ	4	1.00	0.256
$\gamma = 0$	0	0.00	0.465
dephasing	0	0.00	0.442
depolarizing	0	0.00	0.526
no mixer	0	0.00	0.403
Rx only	0	0.00	0.521
ZZ only	0	0.00	0.503

IV. DISCUSSION AND SCOPE

For this reservoir family, useful learning is organized by a transferable operating regime rather than by task-specific tuning—a hypothesis whose credibility comes from variety. Four tasks with little in common, spanning smooth chaos, nonlinear memory, and irregular real-world data, obey the same region, and that region holds when any one task or any one seed is withheld; what survives this much variation is hard to dismiss as a fluke. The reading we draw is deflationary. The useful region is not picked out by Hilbert-space size or by feature diversity but by memory, which suggests that a quantum reservoir succeeds not because its state space is large but because drive, coupling, and dissipation together keep memory and nonlinearity in balance—the principle classical reservoir theory named long ago [3], [7]. This is also why the ablations bite: strip out damping or mixing and the regime does not weaken, it disappears. The boundaries are drawn on purpose. Everything here is simulated, on one architecture family, with measurement treated as free, so finite-shot noise, calibration drift, and readout cost are all absent and could each deform the regime; we therefore claim neither quantum advantage nor hardware efficiency. The claim is smaller and sturdier—that the operating regime is the right object to study, that it transfers, and that memory is how to find it.

V. CONCLUSION

The usefulness of a dissipative quantum reservoir is not a property of its size but of where it is operated. Across tasks and reservoir initializations, strong performance does not wander the control space; it gathers in a single memory-preserving region where input drive, recurrent coupling, and dissipation hold one another in balance. That region transfers, it survives the removal of the very mechanisms that define it only by collapsing, and it can be found in advance by a diagnostic that asks nothing about the task—only whether the reservoir remembers.

The lesson is deflationary in the most productive sense. The resource that makes this quantum reservoir work is not a quantum mystery but the same fading-memory-and-nonlinearity balance that classical reservoir theory identified long ago [3], [7], realized here through physical channels rather than chosen

TABLE I
QRC-ONLY AUDIT TRAIL. HOLDOUT DATA ARE NOT USED TO DEFINE THE OPERATING REGIME.

Item	Value
Tasks and splits	Mackey–Glass, Lorenz, NARMA10: washout/train/validation/holdout = 300/1600/400/400; annual Sunspots: 20/150/50/60. Inputs are scaled to $[-1, 1]$ and targets standardized.
Seeds and ridge	Reservoir seeds 42–51; ridge grid $\{10^{-8}, 10^{-7}, \dots, 10^3\}$ selected on validation.
Phase grid	QRC16 coarse grid over $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$.
Primary band	$B_{20,0.7}$ contains 4 connected points; the medoid is $(\beta/\pi, \lambda/\pi, \gamma) = (0.04, 0.12, 0.12)$. Relaxing to $B_{20,0.6}$ gives 14 points and $B_{30,0.7}$ gives 46 points.
Stress tests	Focused QRC-only ablation maps at $\gamma = 0.12$ compare base amplitude damping against $\gamma = 0$, dephasing, depolarizing, no mixer, Rx-only mixer, and ZZ-only mixer. The readout, tasks, seeds, and ridge protocol are unchanged.
Diagnostics	Memory capacity and intrinsic processing capacities are computed from simulator trajectories and compared to validation rank and top-decile screening retention.

Phase maps transfer under leave-one-task-out conditioning ($\gamma = 0.12$)

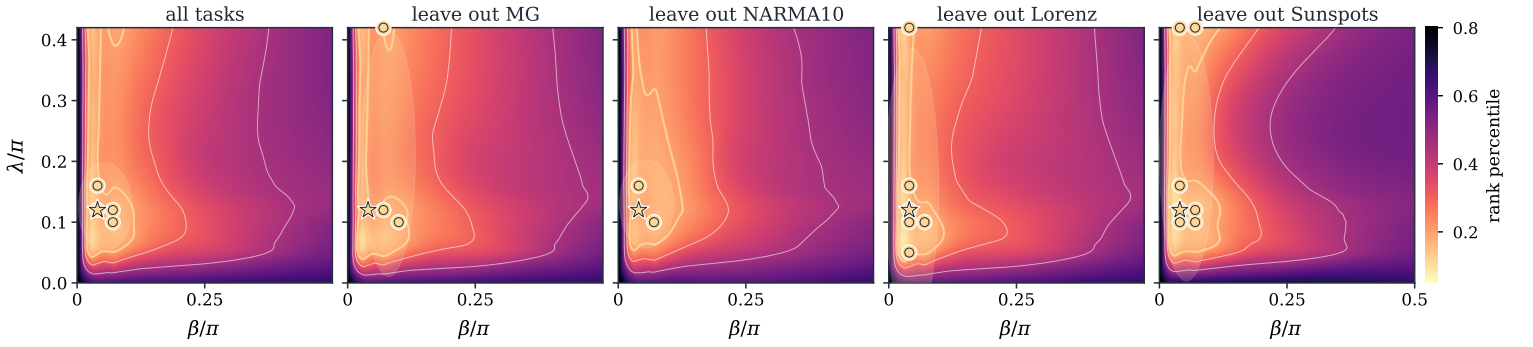


Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. Gold circles mark points that appear in the top-20% for at least 70% of the displayed task-seed replicates; the star is the medoid of the primary band. The same low-input, nonmaximal-coupling region persists under leave-one-task-out conditioning.

Validation-defined band and mechanism-sensitive phase-map geometry

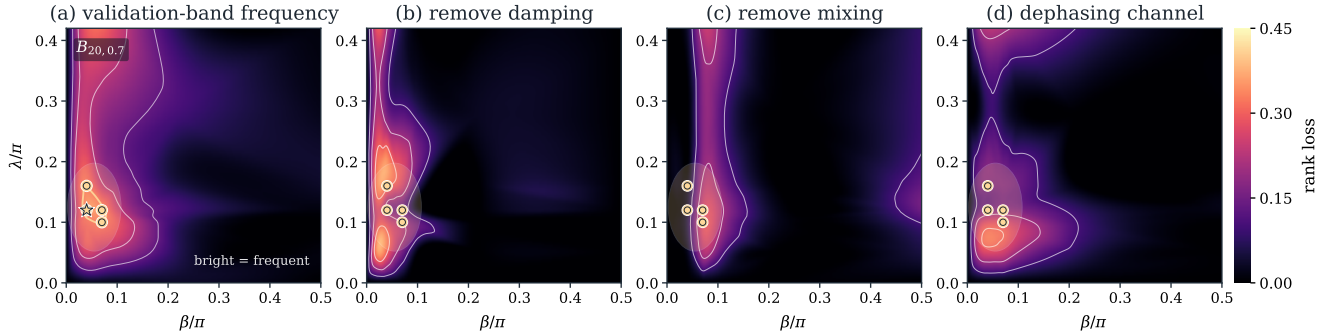


Fig. 2. QRC-only phase-map evidence for the operating-regime hypothesis. (a) The top-20 frequency map visualizes the validation-defined $B_{20,0.7}$ band at $\gamma = 0.12$: bright coordinates repeatedly rank in the top validation quintile across task-seed replicates. (b–d) Ablation rank-loss maps show how much the mean validation rank degrades when controlled damping is removed, recurrent mixing is removed, or amplitude damping is replaced by dephasing. The highlighted base-band projection lies where these mechanism removals cause rank loss.

weights. Read this way, the contribution is methodological before it is physical: it reframes reservoir selection as the task of locating a transferable operating regime, and it offers memory, rather than raw feature diversity, as the compass that

points there.

What remains is the harder half. These results are established in simulation, on a small family of architectures, with measurement treated as free; finite sampling, calibration drift,

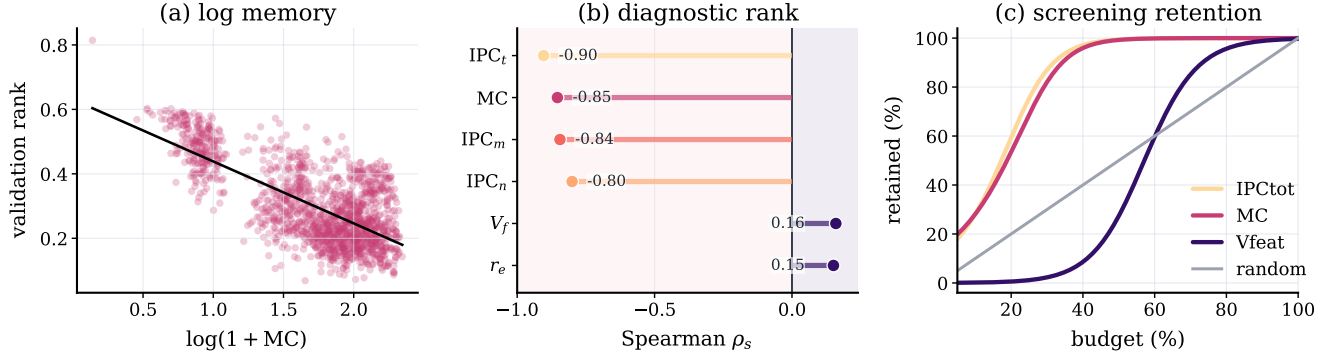


Fig. 3. Diagnostic audit for the QRC phase-map claim. A single log-memory panel expands the non-collapsed memory range and shows the strong monotone relation between retained history and validation rank. Intrinsic processing-capacity diagnostics give the strongest rank correlations, and smooth monotone guides fitted to the empirical screening table show that memory/IPC recovers useful reservoirs earlier than random or diversity-only screens.

TABLE II
HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION.
MEANS ARE OVER TEN SEEDS PER TASK; “ALL” AVERAGES THE 40
TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	8.77e-05	8.19e-05	0.105
Lorenz	5.28e-05	7.36e-05	0.169
NARMA10	0.561	0.520	0.172
Sunspots	0.194	0.179	0.177
All	0.189	0.175	0.156

device noise, and the true cost of reading out many observables all lie outside the present claim. The natural next step is not a larger benchmark but a more honest one—to ask whether the same memory-defined regime persists once the reservoir must be measured, calibrated, and run on hardware that does not forget on command. We expect it to, and we have given the map for finding out.

REPRODUCIBILITY APPENDIX

Frozen reservoir. The simulator uses exact density matrices initialized at $\rho_0 = |0 \dots 0\rangle\langle 0 \dots 0|$ and carried between time steps. For seed s , NumPy `default_rng(s)` draws fixed angles $\theta_i \sim U[0, 2\pi)$ once; the main mixer is $U_{\text{mix}} = \bigotimes_i R_x(\theta_i)$, reused at every time step and layer. The layer unitary is $U_\lambda U_{\text{mix}}$ with ring edges $(i, i + 1 \bmod n)$ and $U_\lambda = \prod_{(i,j)} \exp(-i\lambda Z_i Z_j)$. The exact θ_i values for seeds 42–51 are included in `data/frozen_rx_angles.csv`.

Simulator and features. For $n = 4$, the channel is applied as a column-major superoperator $\sum_k K_k^* \otimes K_k$ after each unitary layer; no finite-shot sampling is used. Amplitude damping uses local Kraus operators $K_0 = \text{diag}(1, \sqrt{1-\gamma})$ and $K_1 = \sqrt{\gamma}|0\rangle\langle 1|$. The main readout concatenates exact expectations of Z_i and ring-pair $Z_i Z_j$ after each of two layers, giving 16 features; ridge regression adds an intercept and does not penalize it.

Diagnostics. Memory and IPC are computed on a common iid diagnostic drive $u_t \sim U[-1, 1]$ of length 900 with washout 150 and generator seed 12345. MC is the sum of held-out R_+^2 for delays 1–10. IPC uses ridge $\alpha = 10^{-8}$, a 70/30 chronological train/test split, memory targets $\sqrt{3}u_{t-d}$ for delays 1–20, quadratic Legendre targets $\sqrt{5}(3u_{t-d}^2 - 1)/2$ for delays 1–10, and ten cross-delay products. Negative R^2 values are clipped to zero.

Uncertainty and artifacts. Confidence intervals are nonparametric bootstrap intervals over the reported held-out transfer values: 50,000 resamples of the mean with replacement, seed 20260523, and 2.5/97.5 percentiles. Code, checked-in CSV/JSON data, generated number macros, and figure artifacts are included with the repository; `./reproduce.sh` rebuilds the QRC-only paper path, and a public archival release will be linked for publication.

REFERENCES

- [1] H. Jaeger and H. Haas, “Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication,” *Science*, vol. 304, no. 5667, pp. 78–80, 2004.
- [2] M. Lukoševičius and H. Jaeger, “Reservoir computing approaches to recurrent neural network training,” *Computer Science Review*, vol. 3, no. 3, pp. 127–149, 2009.
- [3] J. Dambre, D. Verstraeten, B. Schrauwen, and S. Massar, “Information processing capacity of dynamical systems,” *Scientific Reports*, vol. 2, p. 514, 2012.
- [4] K. Fujii and K. Nakajima, “Harnessing disordered-ensemble quantum dynamics for machine learning,” *Physical Review Applied*, vol. 8, p. 024030, 2017.
- [5] S. Ghosh, A. Opala, M. Matuszewski, T. Paterek, and T. C. H. Liew, “Quantum reservoir processing,” *npj Quantum Information*, vol. 5, p. 35, 2019.
- [6] R. Martínez-Peña and J.-P. Ortega, “Quantum reservoir computing in finite dimensions,” *Physical Review E*, vol. 107, p. 035306, 2023.
- [7] G. Tanaka *et al.*, “Recent advances in physical reservoir computing: A review,” *Neural Networks*, vol. 115, pp. 100–123, 2019.
- [8] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, “Dissipation as a resource for quantum reservoir computing,” *Quantum*, vol. 8, p. 1291, 2024.